

PROPERTIES OF SKELETAL MUSCLE

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- **Primary Properties :—**
 - a. **Excitability** : Reaction / response of a tissue to irritation / stimulation.
 - b. **Contractility**
 - c. **Muscle Tone**

1. EXCITABILITY

STRENGTH–DURATION CURVE

- Graph that demonstrates the exact relationship between **strength** and **duration** of a stimulus.

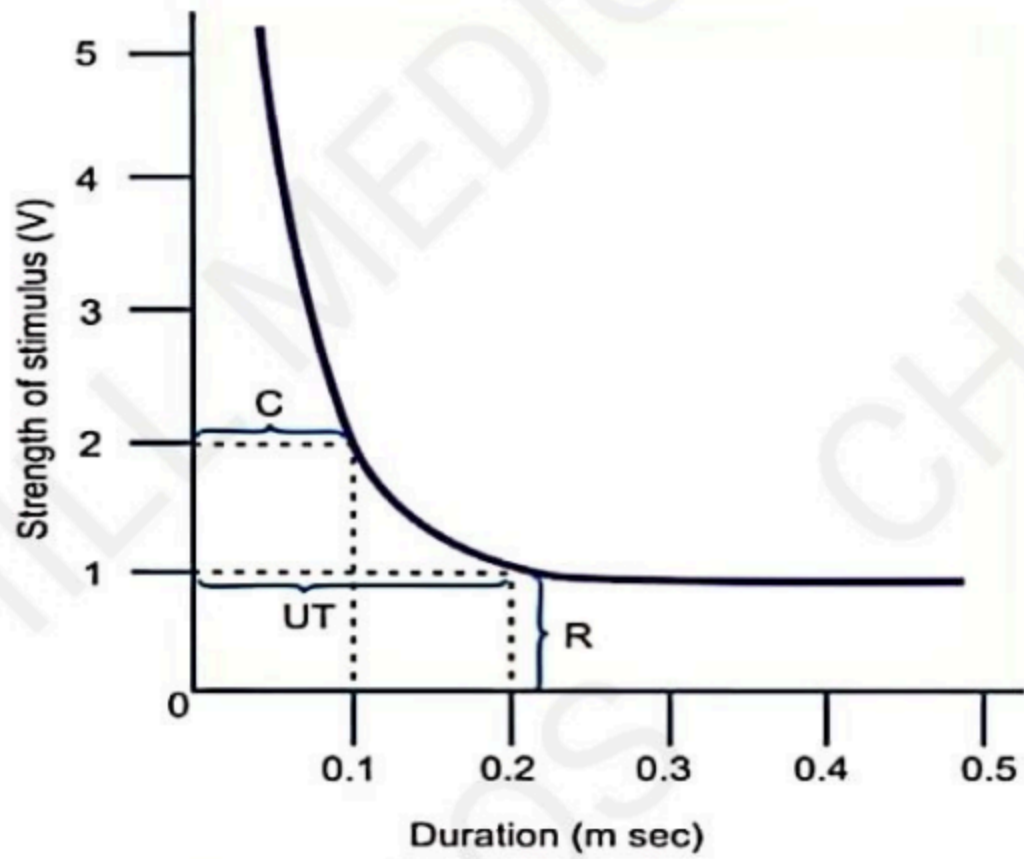


FIGURE 30.1: Strength–duration curve. R = Rheobase, UT = Utilization time, C = Chronaxie.

Characteristic Features of Curve

1. Rheobase (R)

Minimum strength of stimulus required to excite tissue.

2. Utilization Time (UT)

Minimum time required for rheobasic strength of stimulus (threshold strength).

3. Chronaxie (C)

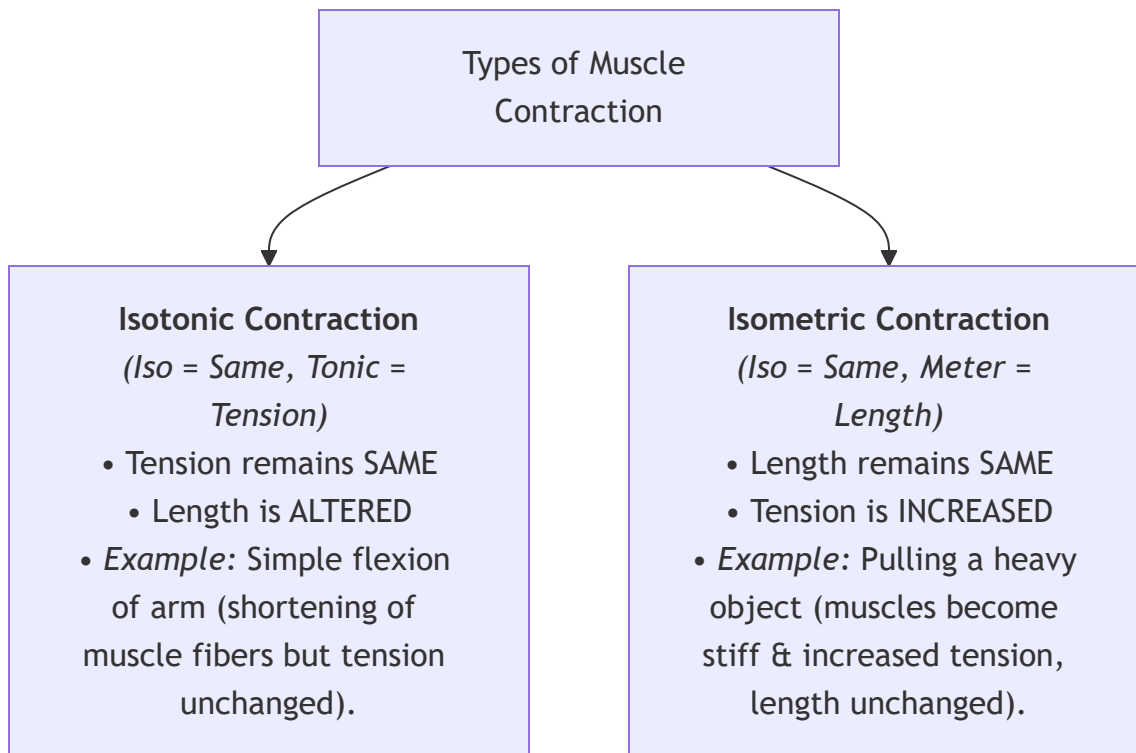
Minimum time required for a stimulus with double the rheobasic strength.

- **Significance of Chronaxie :—**
- Longer the chronaxie, **lesser the excitability**.
- **Values :—**
- In humans = 0.08 – 0.32 msec
- In frogs = 3 msec

2. CONTRACTILITY

- **Response of muscle to a stimulus.**
- **Contraction :** Internal events of muscle with change in either length or tension of muscle fibers.

Types of Contraction



FRANK–STARLING LAW

- States that the **force of contraction is directly proportional to the initial length of muscle fibers.**
- **Effects of Increasing Weight in Afterloaded Condition :—**
 - a. Force of contraction decreases.
 - b. Latent period prolongs.
 - c. Contraction & relaxation periods shorten.

Work done in free-loaded condition is more than in afterloaded condition.

REFRACTORY PERIOD

- **Definition :** Period at which the muscle does not show any response to a stimulus.
- *Reason:* Because one action potential is in progress in the muscle.
- Muscle is unexcitable to further stimulation until it is repolarized.

Classification of Refractory Period:

- a. **Absolute Refractory Period (ARP)** : Period during which muscle does not show any response at all.
- b. **Relative Refractory Period (RRP)** : Period during which muscle shows some response, if strength of stimulus is increased to maximum.

REFRACTORY PERIOD COMPARISON

Feature	Skeletal Muscle	Cardiac Muscle
Duration	Very short (Total = 0.01 sec)	Long (Total = 0.53 sec)
Absolute Refractory Period (ARP)	Falls during 1 st half of latent period (0.005 sec)	Extends throughout contraction period (0.27 sec)
Relative Refractory Period (RRP)	Extends during 2 nd half of latent period (0.005 sec)	Extends during 1 st half of relaxation period (0.26 sec)
Context	Whole of latent period is refractory period	Long compared to skeletal muscle

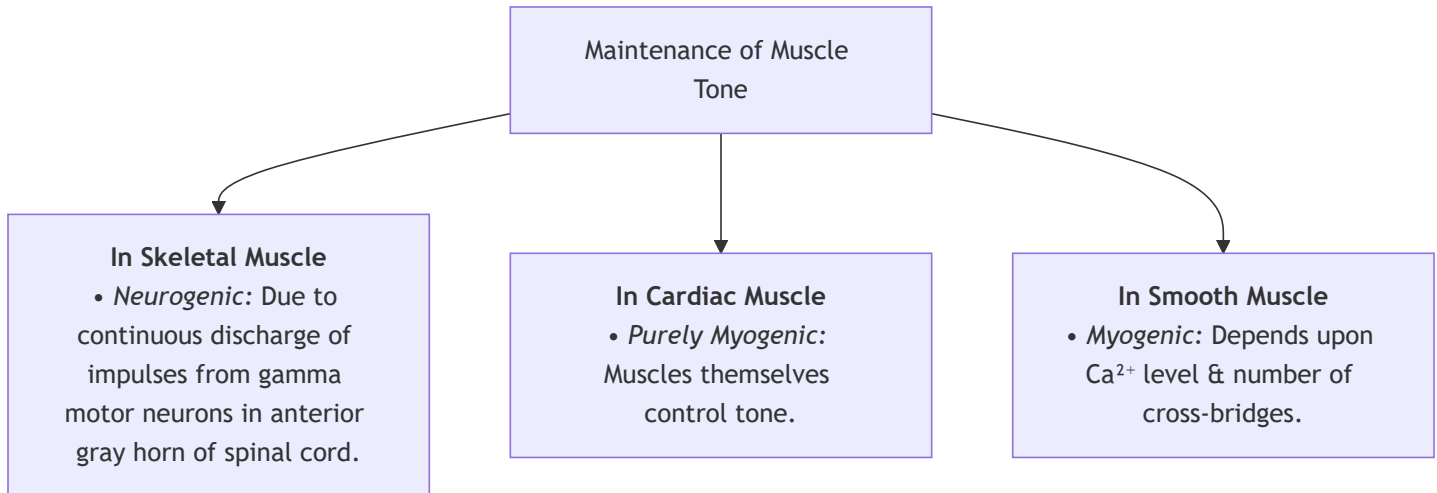
Significance of Long Refractory Period in Cardiac Muscle:

- Cardiac muscle **does not show** :—
- a. Complete summation of contraction
- b. Fatigue
- c. Tetanus

3. MUSCLE TONE

- **Continuous & partial contraction of muscles** with a certain degree of vigor & tension.

Maintenance of Muscle Tone



APPLIED PHYSIOLOGY

(Abnormalities of Muscle Tone)

- a. **Hypertonia** (Increased muscle tone)
- b. **Hypotonia** (Decreased muscle tone)
- c. **Myotonia** (Slow relaxation after contraction)